PROJECTIVE SPACE BUNDLE FORMULA: THE CHOW GROUPS OF A PROJECTIVE BUNDLE ARE FREE OVER THE CHOW GROUPS OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an iterated pushforward computation for injectivity and a decomposition plus induction on the dimension of the base for surjectivity, ending with the reduction to the scheme case

Editorial note. The label lemma-chow-ring-projective-bundle names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The projection formula and the computation of the pushforward of powers of the first Chern class of the tautological quotient by the projection, which makes the universal combination the identity on the base
- Chow groups with rational equivalence for locally finite sums of cycles, and the exact sequences relating the Chow groups of a space, an open subspace and the preimage of its complement
- The reduction of statements about algebraic spaces to schemes along a quasi-compact open, and the scheme case of the projective bundle formula

Source reference. Original lemma 02TX: sources/stacks-chow/tags.txt, source lines 3712–3712. Prerequisite pointer invoked at proof line(s) 4511; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TW: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4321–4321. Prerequisite pointer invoked at proof line(s) 4409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RC: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1306–1306. Prerequisite pointer invoked at proof line(s) 4490; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EEG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof

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line(s) 4438; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TT: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3948–3948. Prerequisite pointer invoked at proof line(s) 4429; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1951–1951. Prerequisite pointer invoked at proof line(s) 4495; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RZ: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1991–1991. Prerequisite pointer invoked at proof line(s) 4470; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 4382; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Projective space bundle formula)**.** *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module $\mathcal{E}$ of rank r. Let $(\pi : P \to X, \mathcal{O}_P(1))$ be the projective bundle associated to $\mathcal{E}$. The map*

$$\bigoplus\nolimits_{i=0}^{r-1} \mathrm{CH}_{k+i}(X) \longrightarrow \mathrm{CH}_{k+r-1}(P),$$

$$(\alpha_0, \ldots, \alpha_{r-1}) \longmapsto \pi^*\alpha_0 + c_1(\mathcal{O}_P(1)) \cap \pi^*\alpha_1 + \ldots + c_1(\mathcal{O}_P(1))^{r-1} \cap \pi^*\alpha_{r-1}$$

is an isomorphism.

Proof. Fix $k \in \mathbf{Z}$. We first show the map is injective. Suppose that $(\alpha_0, \ldots, \alpha_{r-1})$ is an element of the left hand side that maps to zero. By Lemma 02TW we see that

$$0 = \pi_*(\pi^*\alpha_0 + c_1(\mathcal{O}_P(1)) \cap \pi^*\alpha_1 + \ldots + c_1(\mathcal{O}_P(1))^{r-1} \cap \pi^*\alpha_{r-1}) = \alpha_{r-1}$$

Next, we see that

$$0 = \pi_*(c_1(\mathcal{O}_P(1)) \cap (\pi^*\alpha_0 + c_1(\mathcal{O}_P(1)) \cap \pi^*\alpha_1 + \ldots + c_1(\mathcal{O}_P(1))^{r-2} \cap \pi^*\alpha_{r-2})) = \alpha_{r-2}$$

and so on. Hence the map is injective.

To prove the map is surjective, we will argue exactly as in the proof of Lemma 02TT to reduce to the case of schemes. We urge the reader to skip the proof.

Let $\beta \in \mathrm{CH}_{k+r-1}(P)$. Write $\beta = \sum m_j[W_j]$ with $m_j \neq 0$ and W_j pairwise distinct integral closed subspaces of δ-dimension $k + r$. Then the family $\{W_j\}$ is locally finite in P. Let $Z_j \subset X$ be the "image" of W_j as in Lemma 0EEG. For any quasi-compact open $U \subset X$ we see that $\pi^{-1}(U) \cap W_j$ is nonempty only for finitely many j. Hence the collection Z_j of images is a locally finite collection of integral closed subspaces of X.

Consider the fibre product diagrams

$$\begin{array}{ccc} P_j & \longrightarrow & P \\ {\scriptstyle \pi_j}\downarrow & & \downarrow{\scriptstyle \pi} \\ Z_j & \longrightarrow & X \end{array}$$

Suppose that $[W_j] \in Z_{k+r-1}(P_j)$ is rationally equivalent to

$$\pi_j^*\alpha_{j,0} + c_1(\mathcal{O}(1)) \cap \pi_j^*\alpha_{j,1} + \ldots + c_1(\mathcal{O}(1))^{r-1} \cap \pi_j^*\alpha_{j,r-1}$$

for some $(k+i)$-cycle $\alpha_{j,i} \in \mathrm{CH}_{k+i}(Z_j)$. Then $\alpha_i = \sum m_j\beta_{j,i}$ will be a $(k+i)$-cycle on X and

$$\pi^*\alpha_0 + c_1(\mathcal{O}(1)) \cap \pi^*\alpha_1 + \ldots + c_1(\mathcal{O}(1))^{r-1} \cap \pi^*\alpha_{r-1}$$

will be rationally equivalent to β (see Remark 02RZ). This reduces us to the case X integral, and $\alpha = [W]$ for some integral closed subscheme of P dominating X. In particular we may assume that $d = \dim_\delta(X) < \infty$.

Hence we can use induction on $d = \dim_\delta(X)$. If $d < k$, then $\mathrm{CH}_{k+r-1}(X) = 0$ and the lemma holds; this is the base case of the induction. Consider a nonempty open $U \subset X$. Suppose that we can show that

$$\beta|_{\pi^{-1}(U)} = \pi^*\alpha_0 + c_1(\mathcal{O}(1)) \cap \pi^*\alpha_1 + \ldots + c_1(\mathcal{O}(1))^{r-1} \cap \pi^*\alpha_{r-1}$$

for some $\alpha_i \in Z_{k+i}(U)$. By Lemma 02RC we see that $\alpha_i = \alpha_i'|_U$ for some $\alpha_i' \in Z_{k+i}(X)$. By the exact sequences $\mathrm{CH}_{k+i}(\pi^{-1}(X \setminus U)) \to \mathrm{CH}_{k+i}(P) \to \mathrm{CH}_{k+i}(\pi^{-1}(U))$ of Lemma 02RX we see that

$$\beta - \left(\pi^*\alpha_0' + c_1(\mathcal{O}(1)) \cap \pi^*\alpha_1' + \ldots + c_1(\mathcal{O}(1))^{r-1} \cap \pi^*\alpha_{r-1}'\right)$$

comes from a cycle $\beta' \in \mathrm{CH}_{k+r}(\pi^{-1}(X \setminus U))$. Since $\dim_\delta(X \setminus U) < d$ we win by induction on d.

In particular, by replacing X by a suitable open we may assume X is a scheme and we have reduced our problem to Chow Homology, Lemma 02TX. $\square$